MILITARY SPECIFICATION

Projectile, 60MM, ILLUMINATING, M83A3 PARTS, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of mortar projectile designated as Projectile, 60MM, Illuminating, M83A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the Issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105),
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1310

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

FEDERAL

TEST METHOD STANDARD NO. 151 - Metals; Test Methods

DRAWINGS

ARMY

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
9207517 - Projectile, 60MM, Illuminating M83A3, Loading Assembly

PUBLICATIONS

ARMY

ET-9207517 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the Issue in effect on date of Invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 1-199 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Department of Transportation Regulations are a part of the Code of Federal Regulations (1 January 1970 edition) available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the publication should cite "Code of Federal Regulations 49 CFR 1-99 revised as of 1 January 1970.")

ASTM-A-513- Specification for Electric-Welded Steel Tubing

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with all applicable drawings and specifications.

3.2 Heat treatment or stress relief.- Heat treatment or stress relief shall be uniform throughout each lot of body tubes, body tube assemblies, cone tails and tall cone assemblies and shall be so conducted as to produce the physical properties specified on the applicable drawings and in the applicable specifications.

3.3 Physical properties.- Physical properties required by the applicable drawings or specifications shall be uniform throughout each component within the specified tolerances,

3.3.1 Body tubes (See drawing (dwg.) 9223336, 9207531, 9230118 or 9227389).- Welded tubes which are to be used for body tubes shall not rupture, or indicate flaws as evidenced by leaks or undergo permanent deformation when tested as specified in 4.4.7.

3.3.2 Hardness of tall cone (See dwg. 9221406).- The minimum (min.) Rockwell B hardness for the aluminum tail cone shall be 73 for alloy 7075-T6 and 40 for alloy 6061-T6 when determined as specified In paragraph 4.4.4.1.

3.3.3 Carburization of tail cone assembly (See dwg. 9207533).-The depth of carburization of the tail cone (dwg. 9207534) when examined assembled to adapter, fin and coupling shall be .015 min. when determined as specified in 4.4.5.

3.3.4 Pull test for rolled tube, body (See dwg. 9230117).- The rolled body tube shall withstand a two-thousand pound (2,000 lb) min. pull test when tested as specified in paragraph 4.4.3.

3.4 Metal defects .-All metal components shall be free from cracks, splits, cold shuts, inclusions, porosity or other similar defects.

3.5 Loading assembly .-The projectile loading assembly shall comply with all requirements specified on dwg. 9207517 and with all requirements specified in applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

3.6 Moisture

3.6.1 First fire composition.-The moisture content of the first fire composition shall not exceed 0.30 percent, when determined as specified In 4.4.1.1 or 4.4.1.3.

3.6.2 Illuminant composition.-The moisture content of the illuminant composition shall not exceed 0.30 percent, when determined as specified in 4.4.1.2 or 4.4.1.3.

3.7 Burning time.-Candle power of the illuminant assembly (dwg. 9207518).-The illuminant assembly during static testing shall not have a burning time of less than 32 seconds min. The luminosity of all the illuminant assemblies shall not have an average candle power of less than 320,000 adjusted (min.) and no one particular assembly shall have a luminosity of less than 250,000 adjusted (min.) candle power.

3.8 Proving ground

3.8.1 The projectile loading assembly shall comply with the following requirements when tested as specified in 4.4.8. (The requirements of 3.8.2 and 3.8.3 are not applicable when the contractor for the projectile assembly is also the contractor for the assembling and packing of the complete round).

3.8.2 Functioning. -The projectile loading assembly shall function satisfactorily (See 6.8) and shall provide effective illumination (See 6.9) for 32 seconds, min.

3.8.3 Security of parts .-There shall be no metal parts failure in the bore of the weapon or In the first approximate 200 yards of horizontal range.

3.9 Push test of shear pins (dwg. 9207517).-The tops of the shear pins shall be 0.01 inch above flush to 0.01 inch below flush after being subjected to the test specified In 4.4.9.

3.10 Workmanship

3.10.1 All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

3.10.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

3.10.3 Surface finish. -The requirements for surface finish shall be as specified on the applicable drawings.

3.10.4 Protective coatings .-Protective coatings shall be in accordance with the applicable drawing and the specifications listed thereon.

3.10.5 Threads .-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of Specification MIL-A-2550.

3.10.6 Cloth.-All cloth shall be free of tears, rips and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears or foreign matter.

3.11 First article submission.-This specification requires the submission of a first article sample in accordance with the provisions of paragraph 4.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions

4.1.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.2 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced In any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission of parts and assemblies.-The contractor shall submit a first article sample as designated by the contracting officer for determination of conformance to drawing and specification requirements. The first article sample shall consist of the following:

a. Fifteen (15) each of every Internal part and Internal subassembly, Including loaded illuminant loading assemblies.

b. Ten (10) each of every external metal part and every external metal part subassembly unpainted.

c. Five (5) empty projectile loading assemblies pinned and unpainted (i.e., without Internal components).

d, Five (5) empty projectile loading assemblies pinned, painted and marked (i.e., without internal components).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

e. Twenty-five (25) painted and pinned projectile loading assemblies assembled with inert loaded illuminant loading assemblies (suggested filler: sand laminac to simulate candle weight).

f. Fifteen (15) completely loaded projectile loading assemblies, pinned, painted and marked.

The parts and assemblies shall have been produced by the contractor, or furnished by a supplier, using the same production processes, procedures and equipment (except for the inerted candles) as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall Inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of analysis for materials and statements of findings for all inspections and tests performed (See 6.11). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse In production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. -Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. In addition, the twenty-five inert loaded projectile loading assemblies and fifteen loaded illuminant assemblies will be subjected to first article ballistic testing as specified in paragraphs 4.2.3.1 and 4.2.3.2 of this specification.

4.2.3 First Article Ballistic Testing

4.2.3.1 Metal parts security .-Twenty-five (25) of the projectile metal parts shall be tested as shown in the phases in Table I. All of the projectiles shall be Inert loaded to service weight and will be fired from a weapon for which the projectile Is standard. The propellant charge will be adjusted to obtain from 109 to 115 percent (%) of the propellant upper pressure limit at the Charge 4 zone. Dummy or inert fuzes shall be used on all of the metal parts security and recovery test rounds. Pressure data shall be taken for each round fired,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

TABLE I

Phase	Mortar Elevation (Deg)	Initial Production Sample
Security	45%	10
Recovery	-----	15

Recovery.-Rounds fired for recovery which satisfy the requirements of 3.8.3 are for information only to determine quality of ammunition produced. Examination shall be made after recovery for evidence of body or tail cone deformation, dishing or breakup of the projectile other than by impact.

4.2.3.2 Projectile functioning.-After completion of the first article tests as well as the metal parts security test, the projectile functioning tests shall be performed. The fifteen (15) loaded projectiles shall be inspected for conformance to any or all of the requirements of this specification and any or all of the requirements of the applicable drawings at the contractor's plant prior to submission to the proving ground for proving ground testing. Seven of the sample projectiles shall be fired at the Charge 2 zone while the remaining eight will be fired at the Charge 4 zone.

4.2.4 Recovery.-The proving ground shall make the necessary arrangements to recover all of the rounds fired for recovery during the metal parts security test and any or all malfunctioning parachute and illuminant assemblies which may occur during projectile functioning test.

4.2.5 Rejection.-If any internal part, inert or loaded, or external metal parts or assemblies fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. If during the metal parts security or recovery test any projectile fails to comply with the requirement of 3.8.3, the first article sample shall be rejected. In addition, if during the projectile functioning test any loaded projectile fails to comply with the requirements of 3.8.2 or 3.8.3, the first article projectile functioning sample shall be rejected. The Government reserves the right to terminate its inspection and testing upon any failure in the first article sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a complete first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term “inspection lot” as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements, The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified In MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be In accordance with MIL-STD-1168. Each inspection lot of projectile loading assemblies shall contain:

a. Metal parts or metal parts assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and same specification revision.

b. Loaded illuminant canister assemblies from one manufacturer In accordance with the same drawing, same drawing revision, same specification and same specification revision.

c. Parachute assemblies from one manufacturer In accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Heat treatment lots.-Heat treatment lots shall contain components from not more than one heat of steel that has been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of tensile samples as described in paragraph 4.3.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.1.2 Stress relieved lots .-Stress relieved lots shall contain cold worked components from not more than one heat of steel that has been cold worked by the same process, utilizing the same tooling, and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted, shall be the basis for the selection of tensile samples as described in paragraph 4.3.3.5.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated In the applicable paragraph headings. Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent, Sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect. When this method is elected, it shall be applied to all of the Major and Minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 End, cable (See dwg. 9207521, covering a detail of Dwg. 9207517).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter of cable hole (min.)	Gage	01001
Minor:	AQL 2.5 percent		
201.	Flange length	Gage	01002
202.	Length (min.)	Gage	01003
203.	Countersink missing	Visual	01004
204.	Overall outside diameter	Gage	01005
205.	Evidence of poor workmanship (See 3.10)	Visual	01006

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.2 Spacer (See dwg. 9207523, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No. (See 6.2
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.0 percent		
201.	Outside diameter, maximum (max.)	Gage	02001
202.	Thickness	Gage	02002

4.3.2.3 Disc (See dwg. 9207524, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter, min.	Gage	03001
Minor:	AQL 1.5 percent		
201.	Outside diameter, max	Gage	03002
202.	Thickness	Gage	03003
203.	Inside diameter	Gage	03004
204.	Evidence of poor workmanship (See 3.10)	Visual	03005

4.3.2.4 Illuminant suspension assembly (See dwg. 9207519, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Assembly incorrect	Visual	04001
102.	Cable end protruding from sleeve	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	04003

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.5 Case (See dwg. 9207527, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length	Gage	05001
102.	Inside diameter	Gage	05002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	05003

4.3.2.6 Illuminant loading assembly (See dwg. 9207518, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Illuminant suspension assembly improperly assembled or loose	Visual-Manual	06001
102.	First fire slurry missing or disc improperly assembled	Visual	06002
Minor:	AQL 2.5 percent		
201.	Marking incorrect, illegible or missing	Visual	06003
202.	Plug staples missing, not equally spaced nor not flush	Visual	06004
203.	First fire not flush or below flush	Visual	06005
204.	Outside diameter of candle case	Gage	06006
205.	Evidence of poor workmanship (See 3.10)	Visual	06007

4.3.2.7 Tube, Body (See dwg. 9223336, covering a detail of dwg. 9207517)when applicable .

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Major:	AQL 1.0 percent		
101.	Concentricity and perpendicularity of the inside diameter with the fuze adapter threads (2 locations)	Gage	07001
102.	Total length, max.	Gage	07002
103.	Wall thickness, min.	Gage	07003
104.	Inside diameter at tail cone coupling end, average (avg.)	Gage	07004
105.	Pitch diameter, max.	Gage	07005
106.	Minor diameter, max.	Gage	07006
107.	Length of fuze adapter seat, min.	Gage	07007
108.	Roundness of inside diameter at tail cone coupling end	Gage	07008
109.	Roundness of body tube inside diameter (at center)	Gage	07009
110.	Inside body tube diameter(at center), average	Gage	07010
Minor:	AQL 2.5 percent		
201.	Radius, chamfer or notch missing	Visual	07011
202.	Total length, min.	Gage	07012
203.	Surface finish improper	Visual/Tactile	07013
204.	Pitch diameter, min.	Gage	07014
205.	Evidence of poor workmanship (See 3.10)	Visual	07015

4.3.2.8 Tube, body (See dwg. 9207531, covering a detail of dwg. 9207517, when applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length	Gage	08001
102.	Material thickness, min.	Gage	08002
Minor:	AQL 1.0 percent		
201.	Surface finish improper	Visual-Tactile	08003
202.	Evidence of poor workmanship (See 3.10)	Visual	08004

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.9 Adapter (See dwg. 9207532, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Overall outside diameter	Gage	09001
102.	Length of flange	Gage	09002
Minor:	AQL 1.5 percent		
201.	Total length, min.	Gage	09003
202.	Surface finish improper	Visual-Tactile	09004
203.	Staking notch(es) missing	Visual	09005
204.	Evidence of poor workmanship (See 3.10)	Visual	09006

4.3.2.10 Body tube assembly (See dwg. 9207530, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity and perpendicularity of the inside diameter with the fuze adapter threads(2 locations)	Gage	10001
102.	Inside diameter at tail cone coupling end	Gage	10002
103.	Total length, max.	Gage	10003
104.	Pitch diameter, max. (See dwg. 9207532)	Gage	10004
105.	Minor diameter, max. (See dwg. 9207532)	Gage	10005
106.	Roundness of inside diameter at the tail cone coupling end	Gage	10006
107.	Roundness of body tube inside diameter	Gage	10007
108.	Inside body tube diameter (at center)	Gage	10008
Minor:	AQL 1.0 percent		
201.	Pitch diameter, min. (See dwg. 9207532)	Gage	10009
202.	Evidence of poor workmanship (See 3.10)	Visual	10010

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.11 Tail Cone (See dwg. 9207534, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length	Gage	11001
102.	Material thickness, min.	Gage	11002
103.	Metal defective (See 3.4)	Visual	11003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	11004

4.3.2.12 Coupling (See dwg. 9207535, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length	Gage	12001
102.	Length of body tube bearing surface	Gage	12002
103.	Length of flange	Gage	12003
Minor:	AQL 1.0 percent		
201.	Surface finish improper	Visual-Tactile	12004
202.	Evidence of poor workmanship (See 3.10)	Visual	12005

4.3.2.13 Adapter, Fin (See dwg. 9207536, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Total length	Gage	13001
102.	Outside diameter of flange	Gage	13002

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.5 percent		
201.	Chamfer and notch missing	Visual	13003
202.	Surface finish improper	Visual-Tactile	13004
203.	Evidence of poor workmanship (See 3.10)	Visual	13005

4.3.2.14 Tail cone assembly (See dwg. 9207533, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity and perpendicularity of fin adapter threads with body tube assembly bearing surface	Gage	14001
102.	Concentricity of bourrelet with body tube assembly bearing surface	Gage	14002
103.	Diameter of bourrelet	Gage	14003
104.	Diameter of body tube assembly bearing surface	Gage	14004
105.	Total length, max.	Gage	14005
106.	Perpendicularity of coupling shoulder with body tube assembly bearing surface	Gage	14006
107.	Pitch diameter, max. (See dwg. 9207536)	Gage	14007
108.	Minor diameter, max. (See dwg. 9207536)	Gage	14008
Minor:	AQL 1.0 percent		
201.	Pitch diameter, min.	Gage	14009
202.	Evidence of poor workmanship (See 3.10)	Visual	14010

4.3.2.15 Cone, tail (See dwg. 9221406, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Major:	AQL 1.5 percent		
101.	Concentricity and perpendicularity of fin threads with body tube bearing surface	Gage	15001
102.	Perpendicularity of bourrelet shoulder with body tube bearing surface	Gage	15002
103.	Concentricity of bourrelet with body tube bearing surface	Gage	15003
104.	Bourrelet diameter, avg.	Gage	15004
105.	Wall thickness, min.	Gage	15005
106.	Diameter of body tube bearing surface, avg.	Gage	15006
107.	Length of body tube bearing surface, min.	Gage	15007
108.	Pitch diameter, max.	Gage	15008
109.	Minor diameter, max.	Gage	15009
110.	Length of thread to fin adapter face, min.	Gage	15010
111.	Metal defective (See 3.4)	Visual	15011
112.	Roundness of body tube bearing surface	Gage	15012
113.	Roundness of bourrelet diameter	Gage	15013
Minor:	AQL 4.0 percent		
201.	Outside diameter at fin end of tail cone	Gage	15014
202.	Total length	Gage	15015
203.	Diameter of countersink for fin threads	Gage	15016
204.	Pitch diameter, min.	Gage	15017
205.	Length of bourrelet bearing surface	Gage	15018
206.	Diameter directly behind bourrelet diameter, avg.	Gage	15019
207.	Radius or chamfer missing	Visual	15020
208.	Evidence of poor workmanship (See 3.10)	Visual	15021

4.3.2.16 Adapter (See dwg. 9227390, covering a detail of g. 9207517), when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.0 percent		
201.	Chamfer missing	Visual	16001
202.	Evidence of poor workmanship (See 3.10)	Visual	16002

4.3.2.17 Tube, Body (See dwg. 9227389, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length	Gage	17001
102.	Material thickness, min.	Gage	17002
Minor:	AQL 1.5 percent		
201.	Chamfer missing	Visual	17003
202.	Surface finish improper	Visual-Tactile	17004
203.	Evidence of poor workmanship (See 3.10)	Visual	17005

4.3.2.18 Body Tube Assembly (See dwg. 9227391, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity and perpendicularity of the inside diameter with the fuze adapter threads (2 locations)	Gage	18001
102.	Inside diameter at tail cone coupling end, avg.	Gage	18002
103.	Total length, max.	Gage	18003
104.	Pitch diameter, max.	Gage	18004
105.	Minor diameter, max.	Gage	18005
106.	Crimp improper	Visual	18006
107.	Roundness of inside diameter at tail cone coupling end	Gage	18007
108.	Roundness of body tube inside diameter	Gage	18008

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:29Z
Check the source to verify that this is the current version before use.

Minor:	AQL 2.5 percent		
201.	Pitch diameter, min.	Gage	18009
202.	Outside diameter, max.	Gage	18010
203.	Staking notch missing	Visual	18011
204.	Chamfer missing	Visual	18012
205.	Total length, min.	Gage	18013
206.	Evidence of poor workmanship (See 3.10)	Visual	18014

4.3.2.19 Canopy, Parachute (See dwg. 9207529, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Unpatched holes of more than ten (10) broken picks	Visual	19001
Minor:	AQL 1.5 percent		
201.	Width across canopy	Gage	19002
202.	Width of side	Gage	19003
203.	Evidence of poor workmanship (See 3.10)	Visual	19004

4.3.2.20 Parachute Assembly (See dwg. 9207528, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Shroud lines or center cord not properly attached	Visual	20001
102.	Shroud line length from canopy to staple	Gage	20002
103.	One or more shroud lines missing	Visual	20003
104.	Reinforcing tape missing or improperly attached	Visual	20004
105.	Unpatched holes of more than six (6) broken picks	Visual	20005
106.	Shroud line staple missing or not secure	Visual	20006
107.	Improper number of stitches per inch	Gage	20007
108.	Shroud lines crossed or tangled	Visual	20008
109.	Parachute improperly folded	Visual	20009

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.5 percent		
201.	Stitches not extending past reinforcing tape on shroud line	Visual	20010
202.	Center line length from canopy to staple	Gage	20011
203.	Loose, crooked, skipped or improper stitching	Visual	20012
204.	Evidence of poor workmanship (See 3.10)	Visual	20013

4.3.2.21 Body (See dwg. 9224871, covering a detail of dwg. 9207517) when applicable.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter, min.	Gage	21001
102.	Stitching improper	Visual	21002
Minor:	AQL 1.0 percent		
201.	Diameter, max.	Gage	21003
202.	Evidence of poor workmanship (See 3.10)	Visual	21004

4.3.2.22 Disc, Coupling (See dwg. 9207540, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter, min.	Gage	22001
102.	Thickness, min.	Gage	22002
Minor:	AQL 1.0 percent		
201.	Inside diameter, max.	Gage	22003
202.	Evidence of poor workmanship (See 3.10)	Visual	22004

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.23 Parachute Holder Assembly (See dwg. 9224870, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Overall length, min.	Gage	23001
102.	Knot improper	Visual	23002
103.	Improper assembly	Visual	24003
Minor:	AQL 1.0 percent		
201.	Stitching improper	Visual	23004
202.	Evidence of poor workmanship (See 3.10)	Visual	23005

4.3.2.24 Spacer, Parachute (See dwg. 9216686, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length	Gage	24001
102.	Central angle improper	Gage	24002
103.	Material thickness	Gage	24003
104.	Radius improper	Gage	24004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	24005

4.3.2.25 Pin, Shear (See dwg. 9207539, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	25001
102.	Length	Gage	25002
Minor:	AQL 1.0 percent		
201.	Chamfer missing	Visual	25003
202.	Evidence of poor workmanship (See 3.10)	Visual	25004

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.2.26 Parachute Assembly (See dwg. 9207528, sheet 2, covering a detail of dwg. 9207517) prior to insertion into body tube assembly.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Bowline knot improper or not secure	Visual	26001
102.	Bowline knot size or free end improper	Visual	26002
103.	Improper folding (third operation)	Visual	26003
104.	Improper number of folds (fifth operation)	Visual	26004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.10)	Visual	26005

4.3.2.27 Projectile, 60MM, Illuminating, M83A3 Loading Assembly prior to pinning of tail cone assembly (See dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pull cord not threaded through coupling disc with proper knot (See dwg. Note 13)	Visual	27001
102.	Depth of shear pin holes	Gage	27002
Minor:	None defined.		

4.3.2.28 Projectile, 60MM, Illuminating, M83A3, Loading Assembly (See dwg. 9207517).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Shear pin(s) missing	Visual	28001

Source: https://assist.dla.mil -- Downloaded: 20261O-23T00:29Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 percent		
101.	Outside diameter, max.	Gage	28002
102.	Plug loose or insecure	Manual	28003
103.	Improper assembly	Visual	28004
104.	Pin(s) not staked in place	Visual	28005
Minor:	AQL 1.5 percent		
201.	Pin(s) excessively above or below flush	Gage	28006
202.	Marking missing, incorrect or illegible	Visual	28007
203.	Pins improperly located, min.	Visual	28008
204.	Evidence of poor workmanship (See 3.10)	Visual	28009

4.3.2.29 Tube, Body, Rolled (See dwg. 9230117, covering a detail of dwg. 9207517).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity of inside diameter at the tail cone coupling end with pitch diameter of threads including perpendicularity with forward face (2 locations)	Gage	29001
102.	Inside diameter at tail cone coupling end, avg.	Gage	29002
103.	Roundness of inside diameter at tail cone coupling end	Gage	29003
104.	Total length, max.	Gage	29004
105.	Pitch diameter, max.	Gage	29005
106.	Minor diameter, max.	Gage	29006
107.	Excessive gap at rolled end	Gage	29007
108.	Rolled end improper	Visual	29008
109.	Roundness of body tube inside diameter	Gage	29009
110.	Concentricity of the outside diameter at rolled end with pitch diameter of the fuze threads	Gage	29010
111.	Outside diameter, max.	Gage	29011

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Minor:	**AQL 2.5 percent**		
201.	Pitch diameter, min.	Gage	29012
202.	Staking notch missing	Visual	29013
203.	Chamfer missing	Visual	29014
204.	Total length, min.	Gage	29015
205.	Surface finish improper	Visual	29016
206.	Evidence of poor workmanship (See 3.10)	Visual	29017

4.3.3 Testing

4.3.3.1 Moisture content

4.3.3.1.1 First fire composition, Major Defect, Code No. 06008 .-The contractor shall provide adequate controls to insure. compliance with the requirements and shall select a three to four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirements, the eight hours of production represented by the sample shall, be rejected. (See 3.6.1).

4.3.3.1.2 Illuminant composition, Major Defect, Code No. 06009 .-The contractor shall provide adequate controls to Insure compliance with the requirements and shall select a three to four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample, after analysts, fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected. (See 3.6.2).

4.3.3.2 Static functioning (See 3.7 and dwg. 9207518), Major Defect, Code No. 06010.-The test shall be performed as specified in 4.4.2. A sample of four illuminant canister assemblies shall be selected out of each batch of candles (See 6.12). Three canister assemblies shall be tested for static burning time, and. if any burning time is less than 32 seconds, or any candle power is less than 250,000 adjusted, the batch shall be rejected. If the average candle power is less than 320,000 adjusted, the fourth sample shall be tested for static burning time, and if the burning time is less than 32 seconds for the fourth sample, or the average candle power of the four samples is less than 320,000 adjusted, the batch represented shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.3.3 Non-destruct electrical test of body tubes (See dwg. 9223336, 9207531, 9227389 or 9230118 (as applicable)), Critical Defects, Code Nos. 07016, 08005, 17006 or 29018, respectively .-All welded tubes that are to be used for body tubes shall be subjected 100 percent to the non-destructive electrical test cited In ASTM-A-513, paragraph S7 (See 6.4).

4.3.3.4 Hydrostatic test of body tubes (See dwg. 9223336, 9207531, 9227389 or 9230118 (as applicable)), Critical Defects-Code Nos. 07017 08006, 17007 or 29019 , respectively.-All welded tubes that are to be used for body tubes shall be subjected to the hydrostatic test specified In 4.4.7 (See 6.4).

4.3.3.5 Tensile test of body tube or body tube assembly, or body tube, rolled (See dwg. 9223336 9207530, 9227391 or 9230117, Major Defects, Code Nos. 07018, 10011 18015 or 29020, respectively .-Two body tubes or body tube assemblies, as applicable, shall be selected from each lot of body tubes or body tube assemblies and shall be subjected to tensile tests to determine compliance with the yield strength, tensile strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot. The test shall be performed as specified in 4.4.6. (See 6.5).

4.3.3.6 Hardness tests

4.3.3.6.1 Hardness test of tall cones (See dwg. 9221406 or 9207533, Note 5), Major Defect, Code No. 15022 and 14011, respectively .-For every lot of tail cones produced, the sampling plan for this test shall be in accordance with code letter G and AQL 0.40 of MIL-STD-105. The lot shall be rejected if the number of defectives equals or exceeds the rejection number specified. Testing shall be performed as specified in paragraph 4.4.4.1.

4.3.3.6.2 Hardness test of carburized tall cone (See dwg. 9207533, Note 10), Major Defect, Code No. 14012.-Twenty-five assemblies shall be selected from each tail cone assembly lot for this test. If any tail cone assembly in the sample exhibits one or more major defects, the lot of tall cone assemblies shall be rejected. The hardness test shall be performed as specified in paragraph 4.4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.3.7 Depth of carburized case of tail cone assembly (See dwg. 9207533), Major Defect, Code No. 14013.-The 25 assemblies selected in accordance with 4.3.3.6.2 shall be used for this test. Failure of any sample to comply with the applicable requirement shall be cause for rejection. The testing shall be as specified in paragraph 4.4.5.

4.3.3.8 Pull tests.-Thirty-two assemblies shall be tested as cited below for the illuminant suspension assembly. Fifty assemblies shall be tested as cited below for the illuminant loading assembly. Fifty assemblies shall be tested as cited below for the body and tail cone assemblies. If any assembly fails to comply with the minimum pull or push requirement by yielding under load the lot represented by the sample shall be rejected.

4.3.3.8.1 Illuminant suspension assembly (See dwg. 9207519), Major Defect, Code No. 04004.-Non-destructive 200-pound pull test.

4.3.3.8.1.1 Illuminant loading assembly (See dwg. 9207518), Major Defect, Code No. 06011.-Non-destructive pull test shall comply with the applicable requirement on dwg. 9207518.

4.3.3.8.2 Body tube assembly (See dwg. 9207530), Major Defect, Code No. 10012 .-Non-destructive 2000 pound pull test.

4.3.3.8.2.1 Body tube assembly (See dwg. 9227391), Major Defect, Code No. 18016.-Non-destructive 2000 pound pull test.

4.3.3.8.2.2 Tube, body, rolled (See dwg. 9230117), Code No. 29021.-Non-destructive two-thousand pound (2,000 lb.) min. pull test.

4.3.3.8.3 Tail cone assembly (See dwg. 9207533), Major Defect, Code No. 14014.-Non-destructive 800 lb. pull test.

4.3.3.9 Profile and alignment (See dwg. 9207517), Major Defect, Code No. 28010.-The loading assembly shall be gaged 100 percent for profile and alignment. Any loading assembly which fails to comply with the requirement specified shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.3.4. This requirement is not applicable when the contractor for the loading assembly is also the contractor for the complete round.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.3.10 Proving ground (See 3.8).-Defects shall be classified in accordance with the following table: The test shall be performed as specified in 4.4.8 (See 6.7).

Categories	Defects	Code No.
Critical	Premature burst or metal parts separation In first approximate 200 yards horizontal range (See 3.8.3)	28011
Major	Tail cone and body tube assembly fall to separate (See 3.8.2)	28012
Major	Illuminant fails to ignite or provides effective illumination for 25.0 seconds or less (See 3.8.2)	28013
Minor	Effective illumination for more than 25.0 seconds, but less than 32.0 seconds (See 3.8.2)	28014

4.3.3.10.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified herein, a sample of 86 loaded assemblies shall be randomly selected from each lot for this test. If any critical defect occurs; if a total of 6 or more major defects occur or if eight or more minor defects occur, the lot shall be rejected.

4.3.3.10.2 After three consecutive lots have met the critiera of 4.3.3.10.1, a sample of 44 loaded assemblies shall be randomly selected from each lot for this test. If any critical defect occurs; if a total of 4 or more major defects occur or if 6 or more minor defects occur, the lot shall be rejected. In the event that three successive lots fall to meet the requirements of 4.3.3.10.2, the sampling provisions of 4.3.3.10.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein, the provisions of 4.3.3.10.2 shall be reinstated.

4.3.3.11 Push test of shear pins (See 3.9), Major Defect, Code No. 28015.-The sampling plan for this test shall be in accordance with MIL-STD-205, Code Letter M (Single Sampling), with an AQL of .04 percent [sample size 315). If one or more loading assembles fail to comply with push test requirements specified in 3.9, the lot of loading assemblies shall be rejected. When alternative steel tail cone assembly (dwg. 9207533) is utilized 100% inspection shall be in effect. Any loading assembly which fails to comply with the requirement specified in 3.3 shall be classed defective and removed from the lot.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation Number ET-9207517 (See 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.3.4.1 (government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement manufactures testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's Inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of Its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Moisture

4.4.1.1 First fire composition.-The moisture content of the first fire composition shall be determined in accordance with Standard MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of trlchloro ethylene and anhydrous methanol.

4.4.1.2 Illuminant composition.-The moisture content of the illminant composition shall be determined in accordance with the Karl Fischer Method as stated in Standard MIL-STD-1234, Method 101.2, up to paragraph 5.3, A 50 milliliter (ml.) sample of 1:1 dimethylformamidemethyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams (g) Of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

A blank sample of the 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole Into the dry titration beaker containing a plastic-coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.5 ± 0.25 minutes. 25 ml. of the supernatant solution shall be rapidly Introduced, accurately pipetted from the supernatant solution liquid Into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls.) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide-methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation: $\frac{AB}{5\ (10)}$ = _____ Percent H20

Less _____ Std. Blank

Leaves _____ Percent Moisture In Sample

Where:

A = M1s. of Karl Fischer to titrate sample
B = Standardization Factor (mg./ml.)
10 = Constant Factor
5 = Standard Sample Weight

4.4.1.3 Alternate method for first fire composition moisture and illuminant composition moisture.-The moisture content of the first fire composition and illuminant composition shall be determined In accordance with Method T101.4, Standard MIL-STD-286. The temperature setting shall be 100 degrees Centigrade (C) and the time setting 30 minutes. The sample size shall be between 0.25 and 0,50 grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.4.2 Static functioning test.-In calculating the candle power of any one illuminant assembly, the best continuous thirty-two seconds of illuminometer readings (at 5-second Intervals when an automatic recorder is not used) are to be used. The first fire disc assemblies shall be removed prior to testing. The static test shall be conducted In accordance with the procedures and equipment specified In paragraph 4.3.4. The distance from the photometric cell to the flare shell shall be 60.00 ± .25 feet. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back Wall is 15 feet. The value recorded for candlepower for each candle tested shall be adjusted by adding 150,000 candles. The adjusted value shall be used to determine compliance with the requirements specified. The center line of baffle opening shall be in line with axial center of fireball and axial center of photometric cell.

4.4.3 Pull test of assemblies.-The pull test shall be conducted in accordance with the procedure and equipment specified in 4.3.4. Only brazed body tube assemblies (See dwg. 9207530), crimped body tube assemblies (See dwg. 9227391) and rolled body tube assemblies (See dwg. 9230117) passing test may be returned to the lot.

4.4.4 Hardness tests

4.4.4.1 Two Rockwell B hardness readings (approximately 180 degrees apart) shall be taken on the outside surface of the tail cone as shown in 4.3.4. The resultant readings shall be averaged to determine conformance with the requirement specified. Test equipent and procedures used shall be in accordance with Mthd ASTM E18 of Federal Standard 151 and 4.3.4.

4.4.4.2 Hardness readings for carburized tail cones.-The 25 sample assemblies shall be sectioned in half along the axis of the cone. A total of four hardess readings shall be taken per tail cone section on the Internal case only. The locations for the readings are as follows:

a. Two readings are to be taken at distances of approximately one inch arc length from each end of one of the exposed sections and at a distance of not more than 1/2 inch from the edge of the coupling but not on the coupling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

b. One reading shall be taken at the approximate longitudinal middle of the tail cone.

c. One reading shall be taken at the fin adapter end within ½ inch from the end of the adapter, but not over the adapter.

A numerical average of the four readings shall be considered as the hardness of the tail cone. Observation shall be made for compliance with the requirements specified. The test surface of the tail cone shall be clean, dry, free from scale, pits and foreign material (such as copper blush) that might crush or flow under test pressure and so affect the reading. The tall cone section shall be adequately supported by an anvil such that the axis of the penetrator shall be normal to the test surface area. Additional procedures to be used shall be in accordance with Mthd ASTM E18 or Federal Standard 151. Equipment used shall be in accordance with 4.3.4.

4.4.5 Depth of carburized case of tail cone assembly.-The remaining half of the tail cone sections used for testing in accordance with 4.4.4.2 shall be used for this test. Each one of the 25 halved sections shall be prepared for cross section metallurgical examination by polishing and etching one of the exposed cross sections to such a degree that when using a 20X Brinell Glass, the line of demarcation between the case and the cone may be clearly distinguished. A total of six readings per tall cone assembly section shall be taken as follows:

A set of readings (constituting both external and internal case) shall be taken within ¼ inch from the edge of the coupling shoulder, but not on the coupling shoulder.

A set of readings (constituting both external and internal case) shall be taken within ± ¼ inch from the center of the tail cone section.

A set of readings (constituting both external and internal case) shall be taken within ½ inch from the edge of the adapter, but not over the adapter.

The numerical average of the six readings shall be considered as the total case depth for the tail cone. In reading the individual case depth, read all of the darkened zone that indicates a difference in carbon from the uniform core structure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

(NOTE: A recommended polishing and etching procedure is to grind or belt sand the tail cone assembly section and final polish with papers 240, 320, and 400. Then etch with a reagent of 5% nitric acid in ethanol or nitric acid in water, whichever gives the best result with regard to distinguishing with the brinell glass the demarcation between case and core). Observation shall be made for compliance with the requirements specified.

4.4.6 Tensile test.-From each sample selected in accordance with 4.3.3.5, two tension test specimens conforming to the tube type specimen-1 prescribed by Method ASTM E8 of Federal Test Methods Standard No. 151 (except that the B dimension shall be 1 1/2 inches long) shall be taken from the tubing, longitudinally and 180 degrees apart. (Test specimens from welded tubes shall be located approximately 90 degrees from the weld). Yield strength shall be determined by either the offset method or the extension under load method, as prescribed by Method ASTM E8. The limiting set shall be 0.2% (0.002 Inches per inch of gage length). Equipment used shall be in accordance with Federal Test Methods Standard No. 151, Method ASTM E8, and 4.3.4.

4.4.7 Hydrostatic test of body tubes.-All welded tubes that are to be used for body tubes shall withstand the Internal hydrostatic pressure as calculated by the following formula. However, the pressure applied to the tube wall shall be 5,000 pounds per square inch (psi) if the formula parameters dictate the use of pressure greater than this value.

$$P = \frac{1.6\ St}{D}$$

Where:

P = Test pressure in psi
S = Min. specified yield point, psi
T = Specified min. wall thickness, in inches
D = Specified max. outside diameter of tube In inches

Any tube which does not sustain the hydrostatic test pressure for a minimum of 10 seconds or evidences a permanent deformation after hydrostatic test shall be rejected. Equipment used shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.4.8 Proving ground.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The projectile assembly shall be assembled with accepted fuzes, fin assemblies, ignition cartridges, primers and propellant increments, and fired from a 60MM mortar for which the round is standard. One half of the sample cartridges shall be fired with an ignition cartridge and two propellant increments (charge zone two) and one half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (charge zone four) and in accordance with the elevation specified in Table II. Observation shall be made for compliance with the requirements specified for security of parts and "effective illumination" (See 4.3.3.10). In addition, muzzle velocity height of burst and time of fuze functioning shall be recorded for information only for each round tested. Any malfunction occurring during the test shall be reported in the official firing record.

TABLE II

Charge Zone	Mortar Elevation (degrees)
2	68
4	45 1/4

4.4.9 Push test of shear pins.-An axial load of 42 ± 2 pounds shall be applied to the top of each shear pin using equipment in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The projectile loading assembly shall be packed In accordance with the Code of Federal Regulations 49 CFR 1-199 in a manner that will afford adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity.

5.2 Marking.-Each shipping container shall be marked in accordance with dwg. 8796522 for interplant shipment. The FSC number and DOD identification codes are not applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1168.

c. Requirements for the submission of first article samples by the contractor.

d. Concentricity.-Concentricity requirements for Tube, Bodies; Body Tube Assemblies; Cone, Tails and Tail Cone Assemblies may be liberalized or omitted by mutual written agreement between the Government and prime contractor for the projectile assembly provided the contractor agrees to accept the resultant loss of projectile alignment rejects (See 4.3.3.9) as a result of the action.

6.2 Inspection code numbers. -The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment designs.-Inspection equipment designs are of two types: Army designs and contractor designs. Wherever required, Army designs are designated by drawing numbers on the Equipment Lists referenced on the ET, and are mandatory for use by the contractor. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD".

6.3.1 Army designs .-Mandatory Army designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.3.2 Contractor designs. -Contractor designs are required for all inspection equipment for which Army designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified inspection equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by means of catalog sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with Army designs if not otherwise covered thereby. Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number from section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission for approval .-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 If the hydrostatic and non-destructive electrical tests have been performed by either the mill supplier or metal parts vendor, and the tubing has been found to conform to specification requirements, a second hydrostatic or non-destructive electrical test need not be performed.

6.5 If the mill lot of body tubing does not undergo any stress-relieving or heat-treating during fabricating into final body tubes or body tube assemblies (See dwgs. 9223336, 9207530, 9230118, 9227391, as applicable), the specimens for the tensile, yield and elongation tests may be taken from mill tubes from the lot of tubing rather than from the component tube body or body tube assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.6 Static burning time.-Minimum static burning time as referenced herein (See 3.7) is required as a time base for the determination of luminosity and compliance does not in, itself guarantee final ballistic minimum burning time requirement. Structual integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirements and Is attained by proper formulation, blending, consolidation and cure.

6.7 Burst height. -If a projectile functions properly in all respects with the exception that the “effective illumination” time Is less than thirty-two (32) seconds because the illuminant and parachute assembly hit the ground while still burning, and if the burst heights i.e., the height above ground level at which the tail cone separated from the tube body was less than 492 feet (150 meters), the performance of the projectile shall not be considered in evaluating the acceptability of the lot.

6.8 Satisfactory functioning. -Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure “effective illumination."

6.9 Effective illumination.-Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of the first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as “effective illumination", Any occurrences which detract from “effective illumination” as herein defined shall be recorded on the official firing record.

6.10 Proving ground.-The Proving Ground shall make the necessary arrangements to recover all malfunctions which may occur during ballistic test unless otherwise directed by the Procuring Agency. The terrain in the area specified by the ballistic test conditions from the gun sight to down range shall be suitable so that all malfunctions will be accessible for recovery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.11 Statements of findings shall show the specific analytical results of each test or inspection performed as required by the applicable drawing or specification, together with the specified acceptable limits.

6.12 Batch.-A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical operation intended to make the final product substantially uniform. In addition, for the purpose of this specification, a batch of candles shall be defined as that quantity of candles which has been loaded with one batch of illuminant, using the one batch of illuminant for all increments.

6.13 Submission of test data.-In addition to the normal distribution of records, when the projectile is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-A and SMUPA-AD-E-A4, Dover, New Jersey, 07801-.

Custodian:
Army -MU

Preparing Activity
Army -MU

Project Number: 1310-A-037

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.